Lecturer:	**Algebra I**	**Due date:**
Pedro Núñez	Sheet 3	15 October 2026

Exercise 3.1. Let V be a finite-dimensional vector space.

(a) Define the natural map $V \to V^{\vee\vee}$.

(b) Prove that this map is an isomorphism.

Solution. The natural map sends $v \in V$ to the functional

$$\lambda \longmapsto \lambda(v)$$

on $V^{\vee}$. After choosing a basis, this map is identified with the usual identification of a finite-dimensional vector space with its double dual. ∎

Exercise 3.2. Let

$$0 \to A \to B \to C \to 0$$

be a short exact sequence of vector spaces. Use Exercise 3.1 to describe the dual exact sequence.

Solution. Dualizing gives an exact sequence

$$0 \to C^{\vee} \to B^{\vee} \to A^{\vee} \to 0.$$

The reference to Exercise 3.1 checks cross-references in the exercise template. ∎